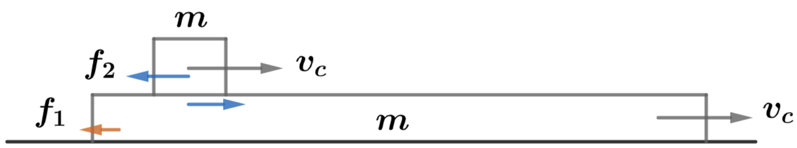
2020B F=ma Exam: Problem 22

Kevin S. Huang



First, note that once the two masses move at the same velocity, they will continue to move together. This is because

$$f_1 = \frac{\mu}{3}(2m)g = \frac{2}{3}\mu mg$$

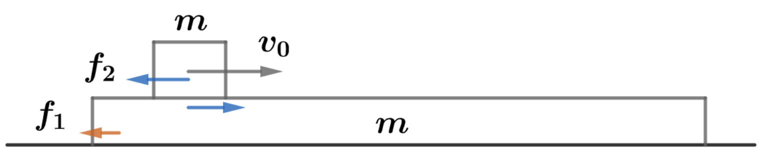
By Newton's 2nd law applied to the whole system,

$$f_1 = (2m)a$$
$$a = \frac{1}{3}\mu g$$

so that implies for the friction on the top block,

$$f_2 = ma = \frac{1}{3}\mu mg < \mu mg$$

which is less than the threshold to overcome static friction. This means we only need to track up to the point when the block is at rest with respect to the slab.



In the beginning when the block starts at velocity v_0, all the friction is kinetic:

$$f_2 = \mu mg$$
$$f_1 = \frac{2}{3}\mu mg$$

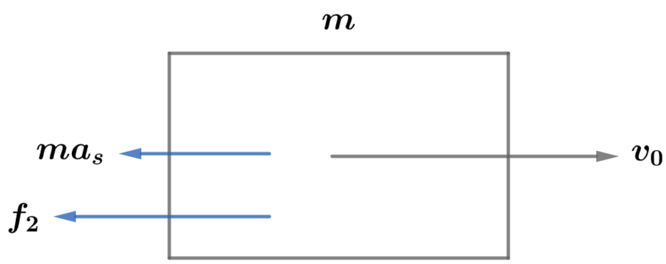
By Newton's 2nd law applied to the slab,

$$f_2 - f_1 = ma_s$$

$$\frac{1}{3}\mu mg = ma_s$$

$$a_s = \frac{1}{3}\mu g$$

accelerating towards the right.



We go to the slab frame and see how far the block slides before going to rest. This is an accelerating frame, so we add a fictitious inertial force ma_s on the block pointing to the left. The acceleration of the block is

$$ma_s + f_2 = ma_b$$

$$\frac{1}{3}\mu mg + \mu mg = ma_b$$

$$a_b = \frac{4}{3}\mu g$$

Recall the kinematics equation,

$$v_f^2 = v_0^2 + 2ax$$

Applied to our case, we have

$$0 = v_0^2 - 2\left(\frac{4}{3}\mu g\right)d$$

$$d = \frac{3v_0^2}{8\mu g} = \frac{3p^2}{8m^2\mu g}$$

using $v_0 = p/m$, so the answer is $\boxed{\text{A}}$.